

Heterolytic fission

Learning outcomes

At the end of this section you should be able to:

- Explain, with equations, the formation of ions by heterolytic fission.
- Explain the relationship between heterolytic fission and nucleophile.

What happens to the electrons when a covalent bond is broken? A covalent bond is formed between a shared pair of electrons. Thus, when the bond breaks, the electrons are supposedly split depending on the type of bond cleavage. Do the electrons stay together on one atom or do each of the electrons go back to their original atoms? In section R3.3.1–2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/radicals-and-how-they-are-produced-id-43489/>), the concept of homolytic fission is introduced: when this bond cleavage occurs each atom receives one of the two electrons that formed the bond. Homolytic fission typically occurs in reactions involving free radicals.

What if the reaction involves charged species as it does in a nucleophilic substitution reaction (section R3.4.1–2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/>)))? This process of bond cleavage occurs differently as one atom receives both the electrons that formed the bond. How does it play a significant role in nucleophilic substitution reactions? How does heterolytic fission work compared to homolytic fission?

When the bond breaks apart and each atom gets one electron, this is homolytic fission. In heterolytic fission, one of the atoms will take both electrons leaving the other atom without any electrons.

Heterolytic fission causes the electrons of a bond to be allocated to one of the species forming two oppositely charged ions – one anion and one cation. Recall that when the covalent bond originally formed, each atom provided one electron. After heterolytic fission, the anion receives both electrons from the bond and therefore will have a negative charge since it contains one more electron than it did before the bond formed. The cation now has one less electron than it did

before the bond was formed, giving it a positive charge after heterolytic fissions (**Figure 1**). The electron pair movement in heterolytic fission is shown using the curly arrow, indicating the movement of a pair of electrons.



Figure 1. Homolytic and heterolytic fission of a covalent bond.

 More information for figure 1

The diagram illustrates two types of covalent bond fission: homolytic and heterolytic. In homolytic fission, denoted at the top, a bond between atoms X and Y splits evenly, with each atom retaining one electron, resulting in two free radicals ($\cdot\text{X}$ and $\cdot\text{Y}$). Curved arrows indicate electron movement. It highlights that this process creates radicals.

In heterolytic fission shown below, the bond between X and Y breaks unevenly, both electrons moving to atom Y. This produces a positively charged ion X^+ and a negatively charged ion $:\text{Y}^-$, represented by respective symbols and curly arrows demonstrating the electron pair movement. The process results in ion formation.

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Making connections

The way in which a bond breaks between two atoms is different in homolytic fission and heterolytic fission. Homolytic fission is covered in [subtopic R3.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43488/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43488/>), a pair of electrons of the bond between two atoms splits equally forming two free radicals that are highly reactive and will participate in chain reactions. When one of these radicals reacts with another molecule, it will produce a new radical.

On the other hand, the uneven splitting of electron pairs of a bond between two atoms in heterolytic fission results in the formation of a new covalent bond in a nucleophilic substitution reaction as it involves the departing of the leaving group. This will produce a new molecule and the ion of the leaving group.

In **Figure 2**, you can see that the breakage of covalent bonds of carbon and chlorine forms cation and anion. The electrons go towards the chlorine atom due to chlorine being more electronegative compared to carbon. This causes chlorine to have a partial negative charge and have a stronger pull on both electrons resulting in a carbocation and a chloride ion.



Figure 2. Heterolytic fission of the carbon—chlorine bond.

 More information for figure 2

The image is a diagram illustrating the process of heterolytic fission of a carbon—chlorine bond. On the left, a carbon (C) and a chlorine (Cl) atom are bonded together. The carbon atom is labeled with a delta positive (δ^+) sign, indicating a partial positive charge, and the chlorine atom is labeled with a delta negative (δ^-) sign, indicating a partial negative charge. Two pairs of dots are shown around the chlorine atom, representing lone pairs of electrons. An arrow curves from the bond towards the chlorine atom, symbolizing the movement of electrons during bond breakage.

To the right of the arrow, the carbon atom is depicted as a cation (C^+), having lost its shared electrons to chlorine, and the chlorine atom is shown as an anion (Cl^-), having gained those electrons. The diagram effectively conveys the shift of electrons towards the more electronegative chlorine resulting in ionic dissociation.

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The process of chlorine leaving chloropropane also can be explained using heterolytic fission. The chlorine atom on chloropropane has a partial negative charge and the carbon atom has a partial positive charge. The hydroxide ion as a nucleophile is attracted to the partially positively charged carbon atom and attacks the carbon resulting in the breaking of the bond between carbon and chlorine. The bonding electrons between these two atoms are transferred to the chlorine atom and cause it to detach from the substrate forming a chloride ion, known as the leaving group.

Figure 3 shows another example of heterolytic fission.



Figure 3. Heterolytic fission of the carbon—hydrogen bond.

 More information for figure 3

The image is a diagram illustrating the heterolytic fission of a carbon-hydrogen bond. On the left, there is a representation of a bond between a carbon (C) atom and a hydrogen (H) atom. The carbon atom has a partial negative charge (δ^-), and the hydrogen atom has a partial positive charge (δ^+). An arrow indicates the direction of electron movement from hydrogen to carbon. On the right side, the resulting ions after the bond breakage are shown: a carbon ion with a full negative charge (C^-) and a hydrogen ion with a full positive charge (H^+). This depicts the shift of bonding electrons to the carbon atom upon bond cleavage, aligning with the difference in electronegativity between carbon and hydrogen.

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Referring to [section S2.2.5 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/) and the electronegativity data that are given in [section 9 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) of the DP chemistry data booklet, the electronegative value of hydrogen is smaller than carbon. Therefore, carbon will have a partial negative charge as it is more electronegative, and hydrogen will have a partial positive charge. When the bond breaks, the electrons will go towards the carbon atom resulting in carbon with a negative charge and hydrogen with a positive charge.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Communication — Collaborate with peers to share ideas, listen actively to other perspectives and encourage others to contribute
- **Time required to complete activity:** 30—40 minutes
- **Activity type:** Group activity

Materials

- modelling clay, at least four different colours, preferably: black for carbon, white for hydrogen, orange for bromine, green for chlorine and red for oxygen
- toothpicks
- paper.

Your task

1. Create two atomic models of the same molecule that contains a bond that will break through homolytic and heterolytic fission.
2. Round up the plasticine like a ball as this will represent the atoms and connect them with toothpicks. When creating the molecules, you should ensure that it contains at least carbon and hydrogen atoms.
3. Choose one person in your group to break one of the bonds on one molecule that can undergo a homolytic fission and ask another person in your group to break another molecule that can undergo a heterolytic fission.
4. Set the broken toothpicks and molecules/atoms such as in **Figure 3** (molecules before and after bond cleavage).
5. Cut the paper to make the curly arrows and place it correctly on both bond cleavages. Round up the plasticine into small balls and place them to show electrons.

Discuss whether your fellow group member has broken the correct bonds on both molecules. Are the bond cleavages set up correctly? Is the direction of the curly arrow correct? How are these two bond cleavages involved in bond breaking of an electron pair? How do these two bond cleavages affect the bond formation? What kind of reactions of the products would form these two bond cleavages?

After that, you can ask another person in your group to show the molecule that has been made and repeat the same process.